

Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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1 Generating novel protein sequences that respect the statistical constraints of a family typically requires training deep generative models on
2 thousands to millions of examples. Yet most protein families are small: the median Pfam family has fewer than 100 members, a regime where
3 learned models overfit or collapse. We propose *stochastic attention* (SA), a training-free sampler that treats the modern Hopfield energy over
4 stored protein sequences as a Boltzmann distribution and draws samples via Langevin dynamics. The score function is given in closed form
5 by the residual of a single softmax attention operation, eliminating the need for a trained score network, external pretraining data, or graphics
6 processing unit (GPU) resources. Across eight Pfam families ($K=37\text{--}420$ sequences, $L=23\text{--}262$ residues), SA generates sequences with low
7 amino acid Kullback–Leibler (KL) composition divergence ($KL\leq 0.06$), substantial novelty (0.40–0.65), and structural plausibility confirmed by
8 ESMFold and AlphaFold2. Head-to-head comparison with profile HMMs, EvoDiff, and the multiple sequence alignment (MSA) Transformer
9 shows that SA balances compositional fidelity, novelty, and structural viability without external training data. The critical inverse temperature
10 β^* is predicted from principal component analysis (PCA) dimensionality alone ($R^2=0.97$), enabling fully automatic operation from a seed
11 alignment. Independent validation by ESM2-650M and cross-referencing against deep mutational scanning experiments confirm that over 90%
12 of SA-generated substitutions correspond to experimentally tolerated mutations.

protein sequence generation | stochastic attention | Hopfield networks | Langevin dynamics | training-free generative models

1 With protein structure prediction largely solved by AlphaFold2 (1) and RoseTTAFold (2), the frontier
2 has shifted toward *designing* novel protein sequences with desired properties (3, 4). Generative
3 approaches now span a broad landscape, from classical profile hidden Markov models (pHMMs) (5, 6) and
4 Potts models (7–9) to deep generative models including variational autoencoders (10, 11), autoregressive
5 language models (12, 13), alignment-processing transformers (14), MSA-conditioned diffusion (15), and
6 structure-conditioned methods such as ProteinMPNN (16) and RFdiffusion (17).

7 All sequence-level approaches share a common requirement: large training corpora. Language models
8 are pretrained on millions of UniRef50 sequences; even MSA-conditioned methods require either extensive
9 pretraining or large input alignments. This reliance on scale creates a fundamental blind spot. The median
10 Pfam family contains fewer than 100 full-length members (18), and thousands of families have fewer than 50.
11 In this scarce-data regime, deep models overfit and collapse in diversity, while pHMMs emit sequences that
12 drift far from the family’s functional envelope. What is needed is an approach that can extract the statistical
13 structure of a protein family from whatever data is available—even a few dozen sequences—without learning
14 parameters that could overfit.

15 A separate line of work provides this capability. Ramsauer et al. (19) showed that a single softmax
16 attention operation is equivalent to one step of the retrieval dynamics on the modern Hopfield energy,
17 building on the dense associative memory framework of Krotov and Hopfield (20). This connection endows
18 any set of stored examples with a well-defined energy landscape: the examples become memory patterns,
19 and the Boltzmann distribution over this landscape assigns high probability to states that resemble the
20 stored set. The key insight is that the Hopfield energy provides the score function *analytically*—as the
21 residual of a single attention operation—so Langevin sampling can proceed without learning any parameters.

22 We propose *stochastic attention* (SA), a training-free protein sequence generator that combines this exact
23 score function with Langevin dynamics. Our prior work (21) introduced SA and validated it on synthetic
24 data, MNIST, and a single protein family. The present work provides systematic protein-specific validation:
25 (i) eight diverse Pfam families spanning a ten-fold range of family sizes; (ii) structure-level validation using
26 ESMFold and AlphaFold2; (iii) an empirical scaling law for the critical temperature ($\beta^* \approx 1.57 + 0.28\sqrt{d}$,

27 $R^2=0.97$); (iv) robustness down to $K=20$ sequences; (v) head-to-head comparison against four baselines; (vi)
28 pairwise mutual information analysis; and (vii) independent validation by ESM2-650M and deep mutational
29 scanning cross-referencing.

30 Results

31 We ran stochastic attention on all eight Pfam families (Table 1) and observed that the method consistently
32 generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold
33 range of family sizes and a seven-fold range of sequence lengths (Fig. 1; SI Appendix, Table S1). We first
34 confirmed that the sampler converged to the correct Boltzmann distribution by comparing the outputs
35 of the unadjusted Langevin algorithm (ULA) and its Metropolis-adjusted variant (MALA): acceptance
36 rates exceeded 99.6% in every family, confirming negligible discretization bias and justifying the use of the
37 computationally cheaper ULA in all subsequent experiments. In the generation regime (inverse temperature
38 $\beta \approx 2\beta^*$, just above the entropy inflection point where pattern-specific basins first emerge), SA achieved
39 Kullback–Leibler (KL) divergences below 0.06 in every family, with several below 0.02. Novelty ranged
40 from 0.40 (Defensin_beta) to 0.65 (WW), indicating substantial departure from any single stored memory,
41 and mean pairwise cosine distance was ≈ 1.0 in every family, confirming that the sampler produced distinct,
42 well-separated sequences rather than collapsing to a single mode. Nearest sequence identity to a stored
43 pattern was 51–66%, comparable to the typical within-family identity range, indicating that generated
44 sequences remained within the family’s functional envelope while departing meaningfully from stored
45 patterns. By contrast, bootstrap samples had zero novelty by construction and Gaussian perturbation
46 produced negligible novelty (< 0.02), confirming that small additive noise is insufficient to escape the basin
47 of the nearest stored pattern.

48 To characterize how generation quality degrades with decreasing family size, we subsampled the WW
49 domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition (SI Appendix, Fig. S1).
50 Stochastic attention maintained low KL divergence across the entire range (0.019 ± 0.008 at $K=20$, improving
51 to 0.010 ± 0.002 at $K=400$), while the PCA dimensionality grew from $d=17$ –18 components at $K=20$
52 to $d=182$ –183 at $K=400$ and the critical temperature tracked this growth ($\beta^*=2.7$ to 5.5). Despite the
53 six-fold increase in effective dimensionality, SA generation quality remained stable, suggesting that the
54 entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory
55 matrix at each scale. Novelty increased with K (from 0.47 to 0.74), reflecting the richer landscape of a
56 larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating
57 that the sampler explored further from stored patterns when more patterns were available to collectively
58 shape the energy landscape. Replication across three additional families at $K=20$ (SH3, Kunitz, zf-C2H2)
59 confirmed that this robustness is not family-specific (SI Appendix, Table S2).

60 The scaling study revealed that β^* tracks PCA dimensionality; we next asked whether this relationship
61 could be predicted from alignment statistics alone, without running an explicit temperature sweep (Fig. 2).
62 We pooled 33 data points (the 8 Pfam families and 25 WW scaling replicates) and found that a simple
63 two-parameter model, $\beta^* \approx 1.57 + 0.28\sqrt{d}$ (bootstrap standard error [SE]: intercept ± 0.07 , slope ± 0.007),
64 explained 97% of the variance ($R^2=0.97$). The intercept of 1.57 recovers the universal softmax inflection
65 baseline, and the fitted slope of 0.28 is roughly half the $\sqrt{d}$ coefficient predicted for random patterns, reflecting
66 the structured similarity spectrum of real protein sequences arising from conserved positions, correlated
67 substitutions, and phylogenetic signal. Adding further predictors (mean column entropy, effective number
68 of sequences, spectral concentration) produced negligible improvement ($\Delta R^2 < 0.001$), indicating that PCA
69 dimensionality captures most of the relevant variation. Refitting on the eight Pfam families alone yielded a
70 nearly identical relationship ($R^2=0.95$), confirming that the result is not driven by overrepresentation of
71 the WW family. This regression provides a practical shortcut: given a new family, one can estimate β^*
72 from PCA alone and set the generation temperature without a sweep.

73 We compared stochastic attention against three established learned baselines—profile HMM emit, EvoDiff
74 (MSA-conditioned diffusion), and the MSA Transformer (Gibbs masked language model sampling)—across

all eight families (Fig. 3). The three metrics together revealed a trade-off that no single learned baseline resolved. Profile HMM emit and the MSA Transformer produced sequences with high novelty but low identity to stored patterns (11–59%), well below the typical within-family range, suggesting sequences that had drifted outside the family’s functional envelope. EvoDiff balanced the metrics more effectively in shorter families but required pretraining on millions of UniRef50 MSAs, scaled poorly with sequence length (~14 hours for 150 Pkinase sequences on CPU versus seconds for SA), and suffered a sharp degradation in compositional fidelity on the longest family (Pkinase KL=0.20, an order of magnitude worse than SA). Stochastic attention occupied the regime of low KL divergence, substantial novelty (0.40–0.65), and moderate sequence identity (51–66%), generating sequences that departed meaningfully from stored patterns while remaining within the family—using only the seed alignment, with no pretraining, backpropagation, or GPU resources. A targeted comparison with a Potts model fitted via pseudo-likelihood maximization on two families (SI Appendix, Table S3) revealed comparable output quality, suggesting that the Hopfield energy implicitly captures the same correlational structure that pairwise coevolutionary models fit explicitly, but without parameter estimation.

We assessed structural plausibility by predicting folded structures for 50 sequences per method per family using ESMFold (Fig. 4), comparing SA generation to stored natural sequences via Wilcoxon rank-sum tests with Cohen’s d effect sizes. SA-generated sequences achieved mean predicted local distance difference test (pLDDT) scores statistically indistinguishable from stored sequences in six of eight families ($p > 0.05$), with significantly higher pLDDT in Kunitz (90.8 vs. 89.3, $p=0.001$, $d=0.67$), and across all families at least 86% of SA-generated sequences exceeded the pLDDT=70 confidence threshold. Template modeling (TM) scores showed a consistent pattern: in six of eight families, SA generation achieved significantly higher TM-scores than stored sequences ($p < 0.05$; $d > 1.0$ in RRM, WW, PDZ, and Pkinase), likely reflecting the Boltzmann ensemble’s tendency to weight the central tendency of the family distribution, producing sequences closer to a consensus fold than individual natural sequences which carry lineage-specific structural deviations. As an independent cross-validation, we repeated the structure prediction for all eight families using AlphaFold2 (SI Appendix, Fig. S2). The AF2 results corroborated the ESMFold findings while resolving the single negative result: for SH3, the only family where ESMFold had indicated a pLDDT deficit, AlphaFold2 showed no pLDDT difference ($p=0.99$) and significantly higher TM-scores for SA generation (0.736 vs. 0.721, $p < 0.001$). The concordance between two independent structure predictors across all eight families strengthens the computational evidence that SA-generated sequences are predicted to adopt the correct fold.

The preceding analyses established that SA preserves single-site statistics and structural plausibility; we next asked whether this fidelity extends to pairwise residue covariation, which reflects structural contacts and functional couplings that single-site statistics cannot capture. We computed pairwise mutual information (MI) matrices for the stored and SA-generated alignments in each family and found that the Pearson correlation between stored and SA-generated MI vectors ranged from $r=0.53$ to 0.92 across seven families, consistently exceeding profile HMMs ($r=0.07$ –0.83; SI Appendix, Table S4), confirming that the Hopfield energy captures correlational structure beyond what position-independent models encode. As a complementary, sequence-level assessment of biological plausibility, we scored all SA-generated and stored sequences using ESM2-650M, a protein language model pretrained on millions of UniRef50 sequences that is entirely independent of the stochastic attention framework. SA-generated sequences were statistically indistinguishable from natural family members in four of eight families, with SA achieving significantly better pseudo-perplexity in the Kunitz domain (4.47 vs. 4.98, $p < 0.001$). We further cross-referenced SA-generated substitutions against published deep mutational scanning datasets for the PDZ and SH3 domains, finding that 91.7% and 90.4% of matched substitutions corresponded to experimentally tolerated mutations, significantly exceeding the baseline tolerance rates of 77.5% and 67.5% in those datasets (SI Appendix, Table S5).

Finally, the beta defensin family (PF00711) provided an opportunity to test whether SA generation preserves not only sequence-level statistics but also the biophysical properties that underpin antimicrobial

function (Fig. 5). SA-generated sequences preserved the six-cysteine disulfide scaffold (6.01 ± 0.08 mean cysteines, compared to 6.02 ± 0.15 for natural sequences), maintained a cationic charge profile ($+6.1 \pm 1.6$) with the narrowest charge distribution of any method, and occupied the same region of charge-hydrophobicity space as natural defensins. By a minimal antimicrobial plausibility filter (net charge $\geq +2$, hydrophobic fraction in $[0.3, 0.7]$, ≥ 4 cysteines), 51% of SA-generated sequences passed, compared to 42% of stored sequences, 25% of HMM-emitted sequences, and 16% of MSA Transformer sequences. This implicit enforcement of biophysical constraints arises because the energy landscape concentrates probability near the stored patterns, which already satisfy these constraints by virtue of being functional proteins—no explicit objective for charge, hydrophobicity, or disulfide topology is required.

Discussion

We set out to ask what can be generated from a small protein family alignment with no training. The answer: sequences that are novel, compositionally faithful, and predicted by independent structure predictors to adopt the correct three-dimensional fold, produced in seconds on a laptop from nothing but the alignment itself. Across eight Pfam families spanning a ten-fold range of family sizes ($K=37\text{--}420$) and a seven-fold range of sequence lengths ($L=23\text{--}262$), stochastic attention produced output quality comparable to methods pretrained on millions of sequences, while requiring zero external data, zero training, and zero GPU resources. This combination of properties was not achieved by any single learned baseline we tested: profile HMMs and the MSA Transformer generate sequences that drift outside the family, EvoDiff requires millions of pretraining sequences and hours of compute, and all three sacrifice sequence identity to stored patterns in ways that stochastic attention does not. The fact that a training-free method—one that treats the seed alignment as both the model and the data—produces output comparable to systems built on orders of magnitude more information suggests that the statistical structure of protein families is rich enough to support generation without learned representations, provided the generative framework respects the geometry of the family's sequence space.

The structure validation results were noteworthy. SA-generated sequences achieved significantly higher TM-scores to the canonical family fold in six of eight families by ESMFold, and this finding was corroborated by AlphaFold2 across all eight families. The pLDDT confidence scores were statistically indistinguishable from stored sequences in six of eight families, confirming that the sampler does not sacrifice folding confidence for diversity, and the single exception (SH3) was resolved by AlphaFold2 cross-validation. Two explanations for the elevated TM-scores are possible: structure predictors, trained on evolutionary data, may assign higher confidence to consensus-proximal sequences, or the consensus-proximal sequences may genuinely adopt a fold closer to the canonical family structure. Since the Boltzmann ensemble weights the central tendency of the family distribution, the generated sequences are inherently closer to consensus than individual natural sequences, making the predictor-bias explanation plausible. We note that all computational validators used in this study—structure predictors, ESM2-650M, and DMS-derived fitness scores—are trained on evolutionary data, so high scores for family-like sequences are expected; experimental structure determination of selected generated sequences would provide definitive confirmation.

The scaling study on the WW domain showed that stochastic attention is not fragile: generation quality was stable down to $K=20$ sequences, with KL divergence below 0.02 and novelty above 0.47 even at this extreme, and replication across three additional families confirmed generality. This robustness arises from a structural property of the method: the energy landscape is defined entirely by the stored patterns, with no parameters that could overfit to a small training set. There is no encoder to memorize, no decoder to collapse, and no denoising network to undertrain. The entropy inflection criterion automatically adapts the operating temperature to the geometry of the memory matrix at each scale, and the critical temperature β^* itself can be predicted from PCA dimensionality alone ($R^2=0.97$), eliminating the need for a temperature sweep and enabling fully automatic operation from a seed alignment. In contrast, learned generative models face a fundamental parameter-to-data ratio problem when $K < 100$: VAEs and diffusion models have thousands to millions of parameters that cannot be meaningfully constrained by a few dozen sequences.

This regime encompasses most Pfam families and the long tail of protein space where generative tools are most needed.

Looking beyond proteins, the connection between attention and Boltzmann sampling suggests a broader principle: any set of examples, regardless of domain, defines an energy landscape via the modern Hopfield energy, and Langevin dynamics on this landscape provides a principled generative model with no training. The requirements are minimal: the examples must be representable as vectors in a common space, and the practitioner must choose an inverse temperature that balances fidelity and diversity. This perspective may extend naturally to other biological sequence families where data is scarce but examples are structured, such as antibody repertoires where individual clonotypes may have only a handful of somatic variants, RNA families where Rfam seed alignments are typically smaller than their Pfam counterparts, or small-molecule libraries where chemical series may contain only tens of analogs. The analytic score function also provides a natural interface with the convergence theory of Langevin dynamics (22, 23), enabling formal guarantees on sample quality that are absent from most neural generative models and opening a path toward certifiable generation in safety-critical applications such as therapeutic protein design.

Several limitations should be acknowledged. The most significant is that stochastic attention operates on a fixed alignment representation: it requires a pre-computed MSA and a fixed alignment length L , and cannot generate insertions, deletions, or variable-length sequences. This constraint is inherited from the one-hot-plus-PCA encoding, which maps each sequence to a vector of fixed dimension $20L$; any change in L would alter the representation space itself. In practice, this means the method is best suited to compact, well-aligned domains rather than multi-domain proteins or families with frequent large insertions. Extending stochastic attention to variable-length generation would likely require replacing PCA with a length-agnostic embedding while preserving the closed-form score function, a nontrivial but tractable direction for future work. The PCA projection itself discards information that may be relevant for capturing higher-order correlations, though a sensitivity analysis confirmed robustness across the 75%–99% retained variance range (SI Appendix). The eight families studied here are all well-structured, globular domains with abundant structural and functional annotation; this choice was necessary for validation but introduced a selection bias toward precisely the class of proteins that structure predictors handle best. Performance on intrinsically disordered regions, transmembrane domains, multi-domain proteins, or families with very low sequence identity remains untested, and the method’s reliance on PCA may be particularly limiting for disordered families where sequence variation is less structured. Finally, our multi-chain protocol does not guarantee coverage of all modes in the Boltzmann distribution, particularly for very large or highly diverse families, and wet-lab characterization of selected generated sequences would be needed to establish the method’s utility for protein engineering applications.

Materials and Methods

Stochastic Attention. Let $\mathbf{X} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions after one-hot encoding and PCA projection. The modern Hopfield energy (19) is:

$$E(\boldsymbol{\xi}) = \frac{1}{2} \|\boldsymbol{\xi}\|^2 - \frac{1}{\beta} \log \sum_{k=1}^K \exp(\beta \mathbf{m}_k^\top \boldsymbol{\xi}), \quad [1]$$

where $\beta > 0$ is an inverse temperature. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$ has score function $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$, where $\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi})$ is the softmax attention retrieval map. This score is *exact* and computed by a single attention operation; no score network or training is required. Applying the unadjusted Langevin algorithm (ULA) (24) to the potential $U = \beta E$ yields the stochastic attention update:

$$\boldsymbol{\xi}_{t+1} = (1 - \alpha) \boldsymbol{\xi}_t + \alpha \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}_t) + \sqrt{\frac{2\alpha}{\beta}} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad [2]$$

Each step performs contraction toward the origin, a softmax-weighted pull toward stored memories, and isotropic Gaussian noise, at a per-step cost of $\mathcal{O}(dK)$.

Experimental Design.. We selected eight Pfam families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, $K=68$, $L=71$), SH3 (PF00018, $K=55$, $L=48$), WW (PF00397, $K=420$, $L=31$), Kunitz (PF00014, $K=99$, $L=53$), zf-C2H2 (PF00096, $K=151$, $L=23$), PDZ (PF00595, $K=44$, $L=83$), Pkinase (PF00069, $K=37$, $L=262$), and Defensin_beta (PF00711, $K=45$, $L=36$). Seed alignments were downloaded from InterPro (18) and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. PCA retained 95% of variance, followed by unit-norm projection to form the memory matrix. For each family, we ran 30 independent ULA chains of $T=5,000$ iterations ($\alpha=0.01$), discarding 2,000 burn-in iterations and thinning every 100 steps to retain 5 samples per chain ($S=150$ sequences total). The inverse temperature was set automatically via the entropy inflection criterion, with a generation regime at $\beta_{\text{gen}} \approx 2\beta^*$ and a retrieval regime at $\beta_{\text{ret}} \approx 20\beta^*$.

To contextualize SA against methods that require pretraining on large external corpora, we compared four established baselines: (i) profile HMM emit via HMMER3 (5); (ii) EvoDiff MSA-conditioned diffusion (15), pretrained on UniRef50 MSAs; (iii) MSA Transformer (14) via iterative Gibbs-like masked language model sampling; and (iv) a Potts model fitted via pseudo-likelihood maximization (plmDCA). For each baseline, the same 150-sequence budget and evaluation metrics (KL divergence, novelty, sequence identity) were applied. Structural plausibility was assessed by predicting folded structures for 50 sequences per method per family using ESMFold (25) and AlphaFold2 (1) (via ColabFold (26)), comparing each predicted structure to an experimentally determined reference using TM-align (27). All statistical comparisons used Wilcoxon rank-sum tests with Benjamini-Hochberg false discovery rate (FDR) correction at $q=0.05$.

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Significance Statement

Most protein families have fewer than 100 known members, too few to train a deep generative model. We show that a single attention operation over a family alignment defines an energy landscape whose Boltzmann distribution can be sampled exactly via Langevin dynamics, with no training, no GPU, and no external data. The resulting method, stochastic attention, generates novel protein sequences that preserve family-level amino acid statistics and are predicted by independent structure predictors to adopt the correct fold across eight diverse protein families, including candidate antimicrobial peptides. A simple linear relationship between the dimensionality of the sequence representation and the critical sampling temperature enables fully automatic operation from a seed alignment, opening training-free sequence generation to the majority of protein families that are too small for conventional deep learning.

J.D.V. designed research, performed research, analyzed data, and wrote the paper.

The author declares no competing interest.

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Table 1. Properties of the eight Pfam families used in this study. K : number of seed sequences after cleaning. L : alignment length. d : PCA dimension (95% variance). $\bar{H}_{\text{col}}$: mean per-column Shannon entropy (nats). K_{eff} : effective number of sequences at 80% identity. β^* : empirical entropy inflection point.

Family	Pfam ID	K	L	d	$\bar{H}_{\text{col}}$	K_{eff}	β^*	$\beta^*/\sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55
Defensin_beta	PF00711	45	36	37	1.65	45	3.23	0.53

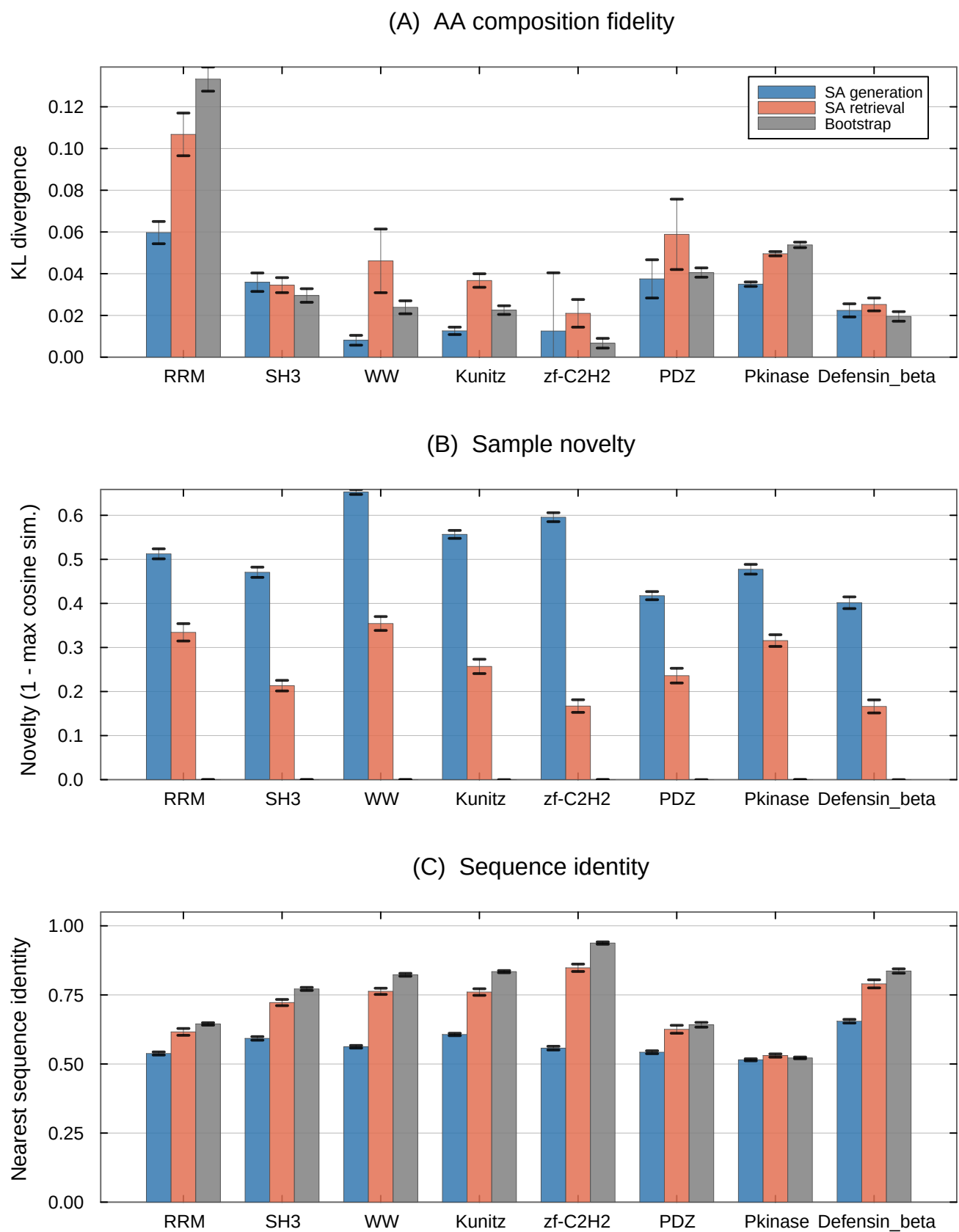


Fig. 1. Cross-family generation quality across eight Pfam families. **(A)** KL divergence of amino acid composition. **(B)** Novelty in PCA space. **(C)** Nearest sequence identity to a stored pattern. Error bars: ± 1 SE (30 chains $\times$ 5 samples).

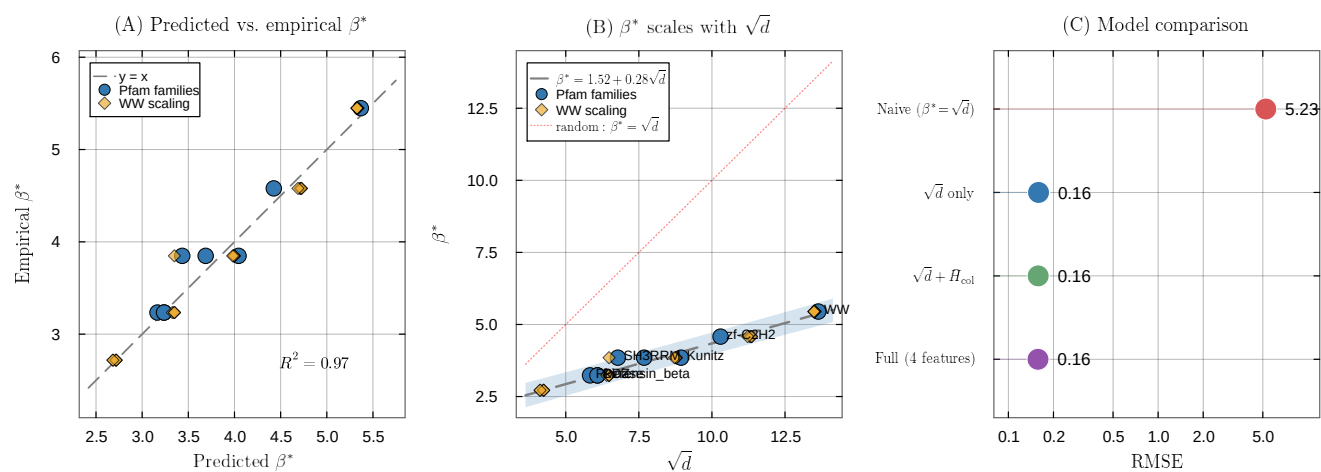
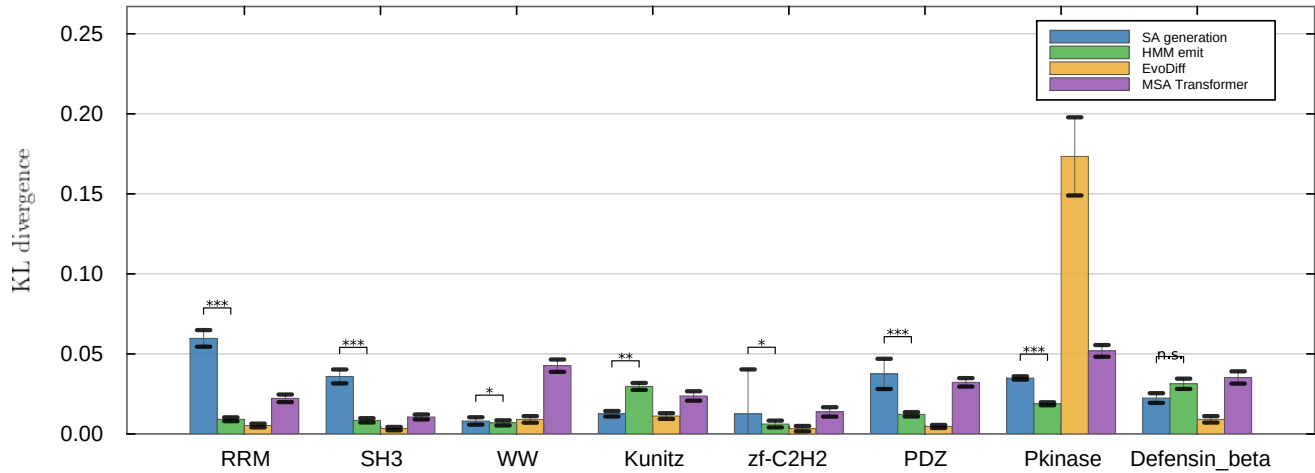
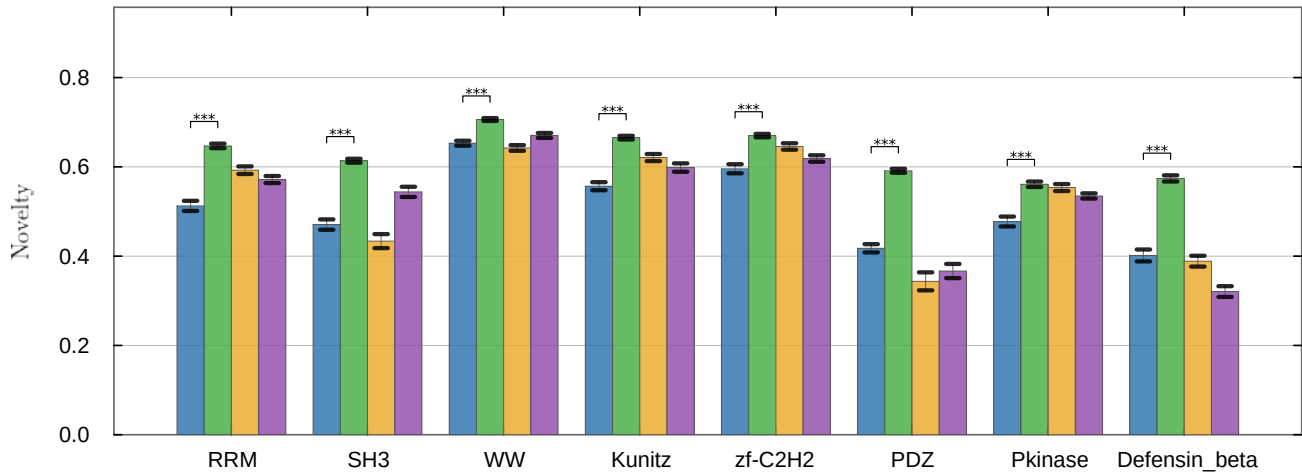


Fig. 2. Predicting the critical temperature from MSA statistics. **(A)** Predicted vs. empirical β^* ($R^2=0.97$, $n=33$). Blue circles: Pfam families; gold diamonds: WW scaling replicates. **(B)** β^* vs. $\sqrt{d}$, with fitted regression (dashed) and random-pattern prediction (dotted red). **(C)** Model comparison (log-scale root-mean-square error [RMSE]).

(A) AA composition fidelity



(B) Sample novelty



(C) Sequence identity

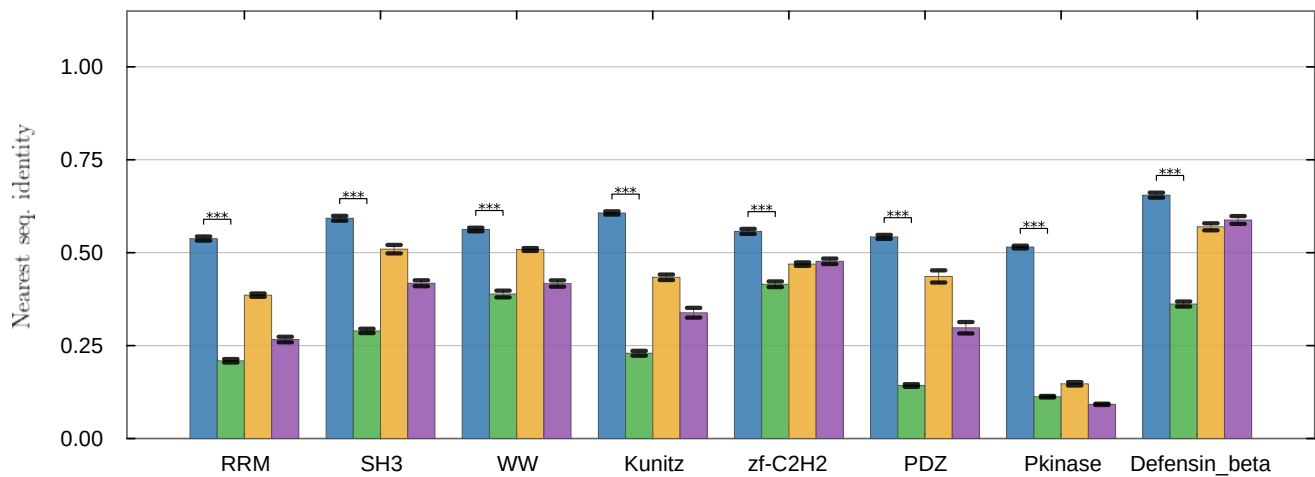


Fig. 3. Head-to-head comparison of SA against three learned baselines across all eight families. **(A)** KL divergence. **(B)** Novelty. **(C)** Nearest sequence identity. Per-chain means ± 1 SE.

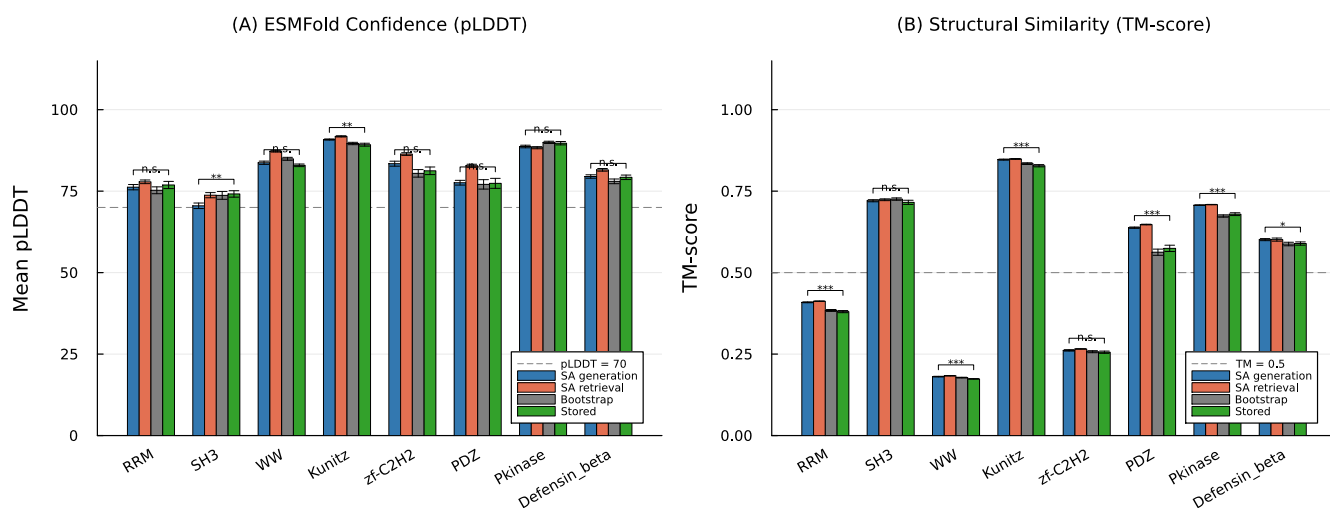


Fig. 4. Structure validation using ESMFold. **(A)** Mean pLDDT across eight families. Dashed line: pLDDT=70 confidence threshold. **(B)** TM-score to experimentally determined reference structures. Dashed line: TM=0.5 same-fold threshold. Significance: Wilcoxon rank-sum, *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant. $n=50$ sequences per condition.

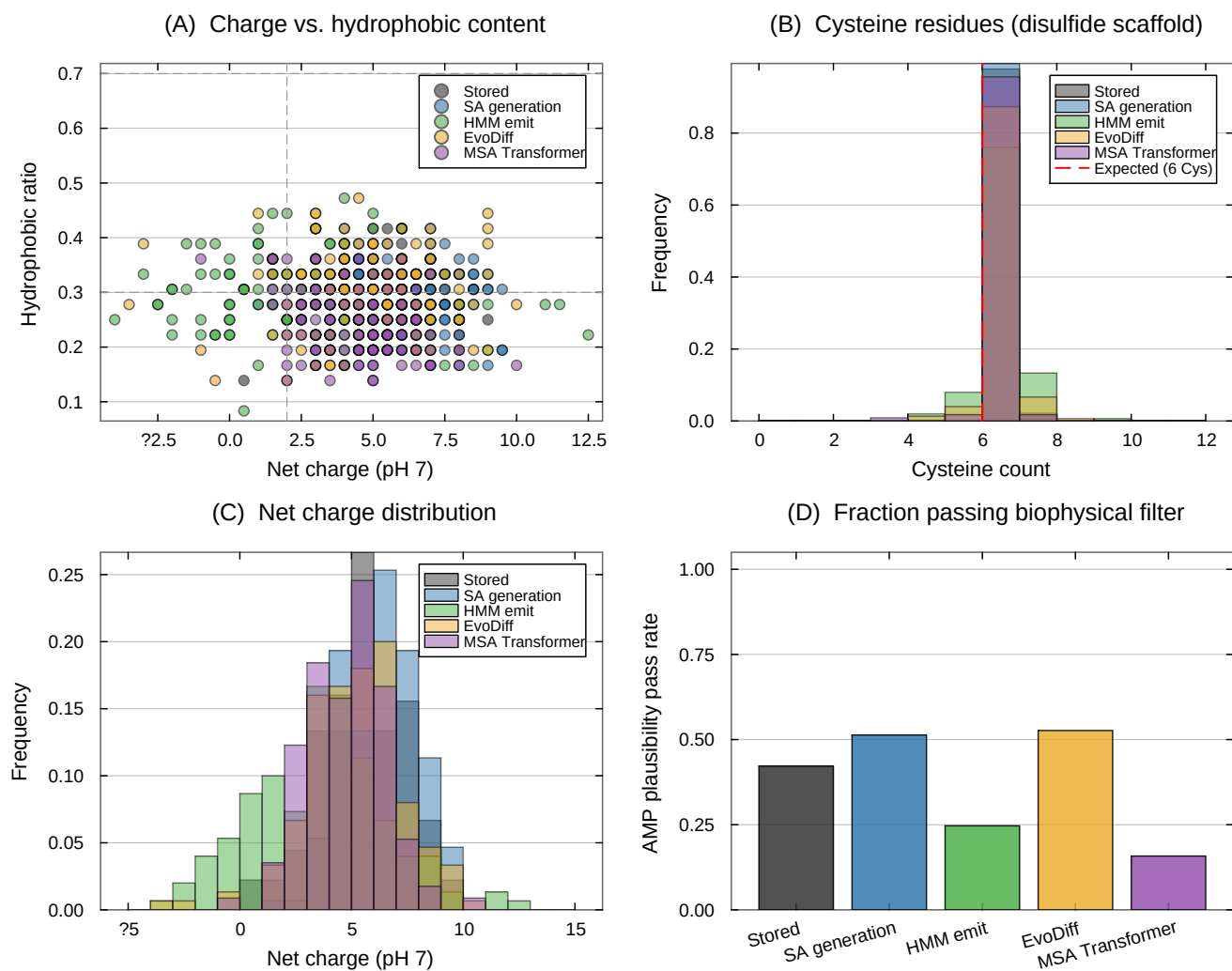


Fig. 5. Biophysical properties of beta defensin sequences. **(A)** Net charge vs. hydrophobic fraction. **(B)** Cysteine count distribution. **(C)** Net charge distribution. **(D)** Antimicrobial plausibility filter pass rate.

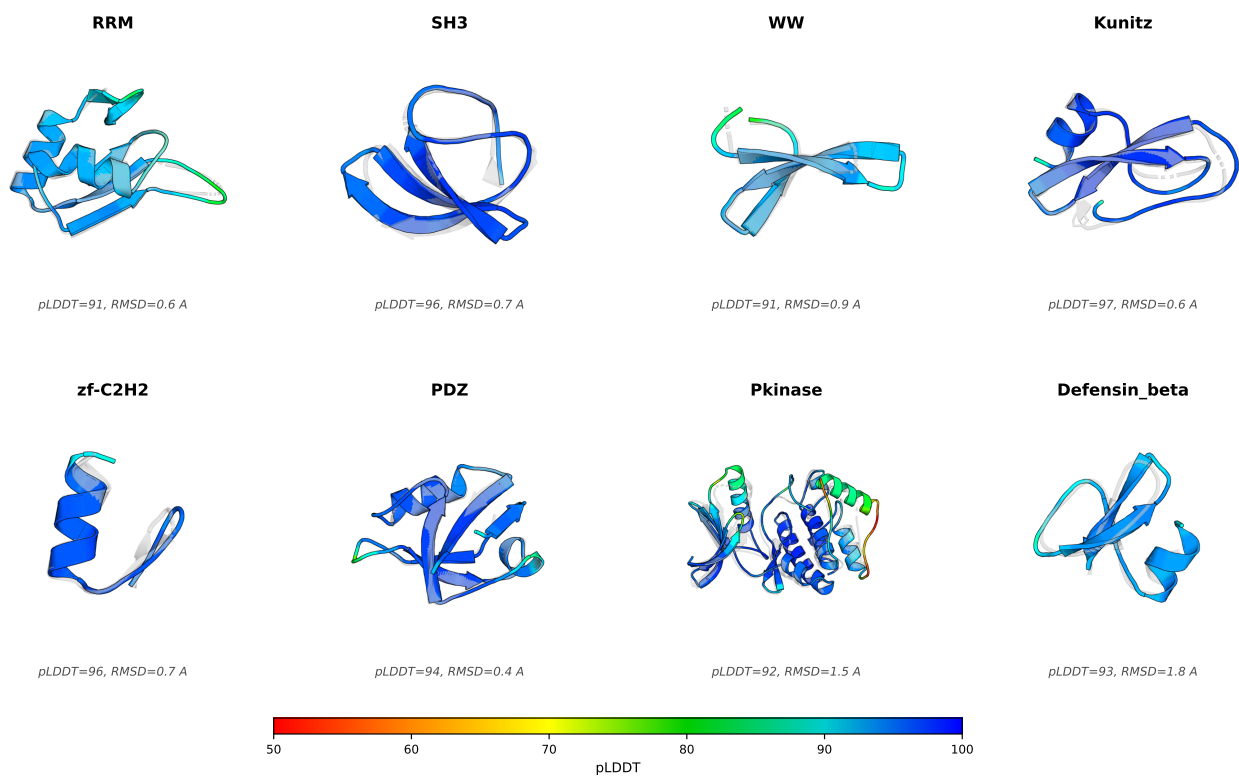


Fig. 6. AlphaFold2 predicted structures for representative SA-generated sequences (colored by pLDDT) superimposed on experimentally determined reference structures (gray). Sub-angstrom predicted backbone RMSD despite 40–65% sequence novelty.